

KeynetikPOS

Feature Documentation

Next-GEN Professional Point of Sale System

Version 2.0.0

Keynetik Solutions • Built with Qt 6 / C++17 • Windows

Overview

KeynetikPOS is a desktop point-of-sale application for retail shops, built with Qt 6 and C++17 on top of an embedded SQLite database. It is designed for the Kenyan market (KSh currency, 16% VAT, M-Pesa-friendly SMS) but is fully configurable for any locale. The application bundles everything a counter needs: a touch-friendly sales screen, multi-cart support, inventory and stock control, role-based staff accounts, receipts, analytics, shift & cash management, automated report messaging, and a licensing/anti-tamper layer.

This document catalogues every feature area of the system, grouped by module.

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1. Point of Sale & Checkout

The main window is the live sales terminal. A searchable, category-filtered grid of product buttons sits beside the active cart and a running order summary.

Sales screen

- Touch-friendly product grid with colour-coded stock status (green healthy, amber low, red critical, grey out-of-stock).
- Live product search box and category drop-down filter.
- Click a product to add it to the active cart; out-of-stock items are blocked.
- Cart table shows item, quantity, unit price and line subtotal with per-item remove.
- Order summary panel: Subtotal, Tax, Discount and a bold Total, updated live.
- Status bar with rolling messages, live date/time clock and current-user label.

Checkout flow

- Checkout opens a payment dialog pre-filled with the amount due.
- Tax is computed from the configurable tax settings on the discounted base (inclusive or exclusive pricing supported).
- **Atomic transaction:** the sale header, every line item and the stock decrement are committed together in a single database transaction — any failure rolls everything back, keeps the cart, and prints no receipt.
- Stock availability is re-validated inside the transaction so concurrent sales cannot oversell.
- On success the receipt prints, inventory warnings refresh, the sale is written to the activity log and the shift, and the cart resets.

Reliability by design

Earlier builds decremented stock twice and printed receipts even on partial failures. Checkout is now a single all-or-nothing operation, so inventory counts and recorded sales always agree.

2. Multi-Cart Management

Cashiers can hold several open transactions at once — useful when a customer steps away to fetch an item or when serving a queue in parallel.

- Tabbed cart bar; each tab is an independent cart with its own items and discount.
- Create a new cart (■ New Cart / Ctrl+N) and switch between carts by tab.
- Rename, close and reorder (movable) tabs.
- Right-click a tab for a context menu to **delete** or **merge** carts.
- Merging combines one cart's items into another.
- Per-cart metadata: created time, last-modified time and live item count.
- New Sale (Ctrl+? / F-key) clears the current cart after confirmation.

3. Discounts

Discounts are applied per cart through a dedicated dialog and are permission- and PIN-gated to prevent unauthorised markdowns.

Capability	Detail
Discount types	Percentage off, fixed amount off, or fixed price; preset quick-discount buttons plus custom entry.
Reason capture	An optional discount reason is stored on the cart and printed on the receipt.
Permission gate	Requires the APPLY_DISCOUNTS permission (Admin, Manager, Cashier).
PIN protection	When enabled, a manager PIN must be entered before a discount is allowed; the PIN is stored as a salted PBKDF2 hash, never plaintext.
Audit	Every applied discount (amount and reason) is written to the user activity log.

4. Payments & Receipts

Payment capture

- Payment dialog records the payment method, amount tendered and computes change.
- Amount paid and change due are persisted with the sale for reconciliation and Z-reports (not just shown on the receipt).
- Payment method is stored per sale and feeds the payment-method analytics.

Receipts

- Three output paths: thermal printer, standard (A4/Letter) printer, and PDF export.
- Receipts carry company header (name, address, phone, tax ID), line items, subtotal, tax, discount & reason, total, payment, change, cashier and a configurable footer.
- Reprint the last receipt from the Sales menu.
- Email-receipt and save-to-PDF hooks for digital copies.
- Optional auto-show of the receipt preview on checkout (setting).

Configurable output

Company details and the receipt footer come from Settings, so receipts rebrand instantly without code changes.

5. Products & Inventory

A full inventory management dialog (tabbed) covers the product catalogue, stock levels, alerts and inventory analytics.

- Product catalogue with name, category, selling price, cost price, profit margin, stock quantity, barcode and active/inactive flag.
- Add, edit and (soft) delete products; selling price can be auto-derived from cost price + margin.
- Category management; products are grouped and filterable by category.
- **CSV import** of products with an imported / skipped / errors summary.
- **CSV export** of the full inventory and of alert lists.
- Inventory cache with auto-refresh on a timer for responsive stock display.
- Low-stock and critical-stock alert views.

Stock status thresholds

Status	Rule	Indicator
Out of stock	quantity = 0	Grey
Critical	1 – 4 units	Red
Low	5 – 20 units	Amber
Healthy	above 20 units	Green

6. Stock Control

A dedicated Stock Manager handles day-to-day stock movements with a full audit trail.

- Searchable, filterable stock table (all / low / critical / out-of-stock).
- Restock products (with optional supplier) and make manual stock adjustments.
- Every adjustment is logged transactionally with old qty, new qty, change, reason and the user who made it — viewable as stock history per product.
- Summary panel: total products, total units, low/critical/out-of-stock counts and total stock value.
- Restocking and selling emit live low/critical/out-of-stock notifications.
- CSV export of stock data.

7. Barcode Scanning

Barcode input is handled by a flexible reader supporting the two common scanner types, configurable from Settings.

Mode	Detail
USB / keyboard-wedge	Scanner types characters + Enter; an application-wide event filter captures the burst and assembles the barcode with an inter-character timeout.
Serial (RS-232 / COM)	Configurable port name and baud rate (default 9600); lists available serial ports on the system.
Cleaning	Configurable prefix/suffix stripping (e.g. STX/ETX framing bytes).
Lookup	A scanned barcode resolves to a product and adds it to the active cart.

8. Users, Roles & Permissions

Access is governed by role-based access control. Login is required at startup (max three attempts) and every privileged action is permission-checked.

Roles

Role	Access summary
Admin	Full access to every feature, including user management, settings, backup and logs.
Manager	Sales, discounts, void; inventory add/edit and stock adjust; all reports & analytics with export; view users.
Cashier	Make & view sales, apply discounts, view inventory.
Viewer	Read-only: view sales, inventory, reports and analytics.

User management

- Create, edit, enable/disable and delete users; deleting a user also clears their activity log in the same transaction.
- Per-user fields: username, full name, email, role, active flag, created-by and last-login.
- Username and email validation; live password-strength meter (Weak/Medium/Strong).
- Users change their own password; admins can reset others'.
- Full activity log: logins, logouts, sales, discounts, user and settings changes.

9. Shift & Cash Management

Shifts tie sales to a cashier and a cash drawer for end-of-day reconciliation.

- Open a shift with a cashier name and an opening cash float.
 - Sales, discounts and voids are recorded against the open shift as they happen.
 - Close a shift with a counted closing float and optional notes.
 - Shift history retrievable (default last 30 days) with per-shift detail.
 - Drives the Z-Report's expected-cash and cash-variance figures.
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10. Reports

Sales reports dialog

- Sales by Date Range
- Sales by Category
- Sales by Payment Method
- Top Selling Products
- Daily Sales Summary

Reports can be exported to **CSV** or **PDF**. A one-click Daily Report is available from the Reports menu.

Z-Report (end of day / shift)

- Gross sales, total discounts, net sales, tax collected and transaction count.
- Breakdown by category and a top-products list.
- Cash reconciliation: opening float, closing float, expected cash and variance.
- Generated per date and/or per shift; exportable and printable.
- Reusable text generator so the same Z-Report can be sent automatically (see §12).

11. Analytics Dashboard

A visual analytics dashboard turns sales data into charts over a selectable date range, plus an at-a-glance quick-stats strip for the main window.

Widget	Shows
Sales trend	Daily revenue over the range with total, average and trend direction.
Category breakdown	Share of sales by category (pie).
Hourly heatmap	Sales intensity by hour of day to spot peak trading times.
Top products	Ranked products by quantity, revenue and profit.
Payment analysis	Split of takings across payment methods.
Customer analytics	Total / new / returning customers, average value, top tier.
Quick stats	Today's sales, transactions, customers and average ticket.

- Selectable start/end date range with refresh-all.
- Export the analytics report to CSV and print it.
- Actual gross-profit reporting (uses stored cost prices) for today, this month or any range.

12. Automated Messaging & Schedules

KeynetikPOS can send reports automatically to staff or owners over SMS / WhatsApp, driven by a scheduling engine with a pluggable provider registry.

Providers

Provider	Use
Africa's Talking	SMS for the Kenyan market (~KES 0.80/SMS, M-Pesa friendly); API key, username and optional sender ID.
Twilio WhatsApp	WhatsApp messages via Twilio (account SID, auth token, WhatsApp-enabled number).

Scheduling engine

- Schedule types: Daily, Weekly, Monthly, On-Shift-Close, On-Sales-Threshold, Custom.
- Report types: Z-Report, X-Report, Inventory Alert, Sales Summary, Custom Report.
- Per-schedule send time, day-of-week / day-of-month, recipient list and provider.
- Active/inactive toggle, next-scheduled tracking, last-sent and failed-attempt counters.
- Timer-driven dispatch with a manual 'send now' for testing.
- Schedules persist in SQLite; managed through schedule and schedule-editor dialogs.
- Delivery and error signals surface success/failure of each send.

13. Refunds

Refunds are processed from the Sales History screen with safeguards against double-refunding.

- Browse sales history; sales already refunded are flagged and locked.
- Process a refund with a captured reason and the processing user recorded.
- Refund is written transactionally and returns the items to stock with logged stock adjustments.
- Sales history is searchable and exportable to CSV; last receipt reprintable.

14. Settings & Configuration

A central settings dialog persists configuration to the database (and provider credentials to user settings).

Group	Settings
Business	Business name, address, phone, email, website.
Currency	Currency symbol and code (defaults to KSh).
Tax	Enable/disable, rate, label (VAT/GST/...), tax-inclusive vs exclusive.
Receipt	Footer text, auto-show receipt preview, show-logo flag.
Discount	Require-PIN toggle and PIN (stored hashed).
Messaging	Twilio SID/token/number; Africa's Talking key/username/sender ID.
Barcode	Input mode, serial port and baud rate.
Theme	Light / dark mode toggle, persisted.

Tax and theme changes propagate live — totals, the tax label and the UI update immediately when settings are saved.

15. Licensing, Trial & Security

Trial & licensing

- 30-day free trial from first run, with reminder prompts and a key-entry prompt on expiry.
- CD-key activation: local activation, or server-backed online activation with device binding.
- Online activations re-validate via a background heartbeat after the window opens — startup never blocks on the network.
- Offline grace window (7 days) for server-backed keys; server-side revocation is honoured at next launch (no mid-shift logout).
- Trial clock is anchored to the earliest of a registry stamp and a signed, machine-bound breadcrumb file, so wiping the registry alone does not reset it.

Integrity & hardening

- Tamper detection: license data is covered by a salted integrity hash (key, install date, activation source and revocation flag); deleting or editing any of it locks the app.
- Passwords and the discount PIN use salted PBKDF2-SHA256; legacy SHA-256 hashes are transparently upgraded on next login.
- Forced password change on first login for the default admin and any admin-reset account.
- Minimal release-only anti-debug check (debugger-present), tuned to avoid false positives on legitimate machines.
- All database access uses parameterised queries; SQLite foreign keys enforced.

Default credentials

First run seeds an **admin / admin123** account flagged must-change-password — the app forces a new password before it can be used.

16. Platform, Data & Keyboard Shortcuts

Platform & data

- Qt 6 / C++17 desktop application, packaged for Windows (NSIS installer via CPack, windeployqt bundling, desktop & Start-menu shortcuts).
- Embedded SQLite database stored in the per-user application-data folder; tables auto-created and migrated on launch.
- Sample catalogue (categories & products) seeded on a fresh database.
- Light and dark themes; localised currency formatting.

Keyboard shortcuts

Shortcut	Action
F2	Checkout
F3	Focus product search
F4	Apply discount
F5	New cart
Ctrl+N	New cart
Ctrl+W	Clear cart
Ctrl+L	Logout
Ctrl+Q	Exit

KeynetikPOS v2.0.0 — Keynetik Solutions. This document was generated from the application source and reflects the feature set at the time of writing.